

Next-cancer prediction from first diagnosis · 84k patients · Taiwan 2003–2020

Validation cohort: 810 multi-cancer patients (20% held-out)

Accuracy — first-cancer-only context:
 $R@1 = 0.232$ · $R@3 = 0.497$ · $R@5 = 0.664$

Lead time (time from first to second cancer):
Median = 681 days (1.9 yr) · IQR = 48–1813 days

UADT validation: 32% of C12/C13→C15 transitions occur ≤ 6 months
(consistent with guideline-based 6-month endoscopy)

Timing windows calibrated from 519 observed site-pair transitions in training set

Methodology

Clinical framing:

A patient presents with cancer at site A on date T_0 .

The model receives [CLS, A, MASK] and predicts the site token at MASK.

Only the index cancer is given as context — no future information used.

Why first-cancer-only context?

- Clinically realistic: at first diagnosis, no subsequent cancers are known.
- Tests whether the Transformer has learned a meaningful site-to-site co-occurrence prior beyond static frequency.
- $R@1=0.232$ vs leave-one-out $R@1=0.312$ — the gap quantifies what is learned from temporal sequence context (Scripts 07-08).

How timing windows are generated:

- From training-set multi-cancer patients, compute actual day-gaps between consecutive cancer pairs (first→second, second→third, ...).
- For each (A→B) pair: report empirical Q1/median/Q3.
- Report: 'Recommend surveillance for [B] at [Q1-Q3] days after [A] diagnosis.'
- For pairs not seen in training: use overall median/IQR.

Validation design:

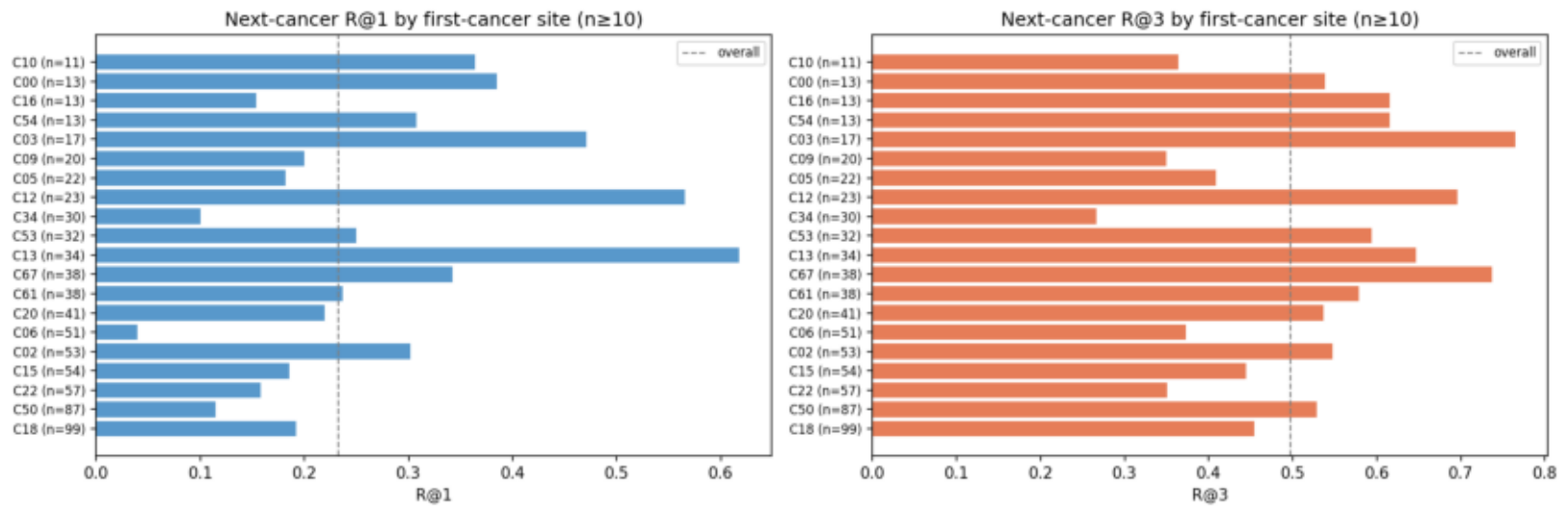
- Same 20% patient-level val split as Transformer training (SEED=42).
- Evaluate: does top-1/3/5 predicted site match actual second cancer?
- Lead time: actual gap days — this is the window in which surveillance would have an opportunity to detect the second cancer.

Comparison to UADT guideline:

Current guidelines (Muto 2004, Huang 2015) recommend synchronous esophagoscopy at HNC diagnosis. The model timing window is compared against this 6-month anchor for C12/C13→C15 pairs.

Surveillance Accuracy by Primary Site

Fig 1: Next-cancer R@1 and R@3 by first-cancer site (overall R@1=0.232, R@3=0.497)
Surveillance accuracy by primary site



Lead Time and Guideline Comparison

Fig 2a: Lead time distribution
(median=681d = 1.9yr, 32% within 6 months)

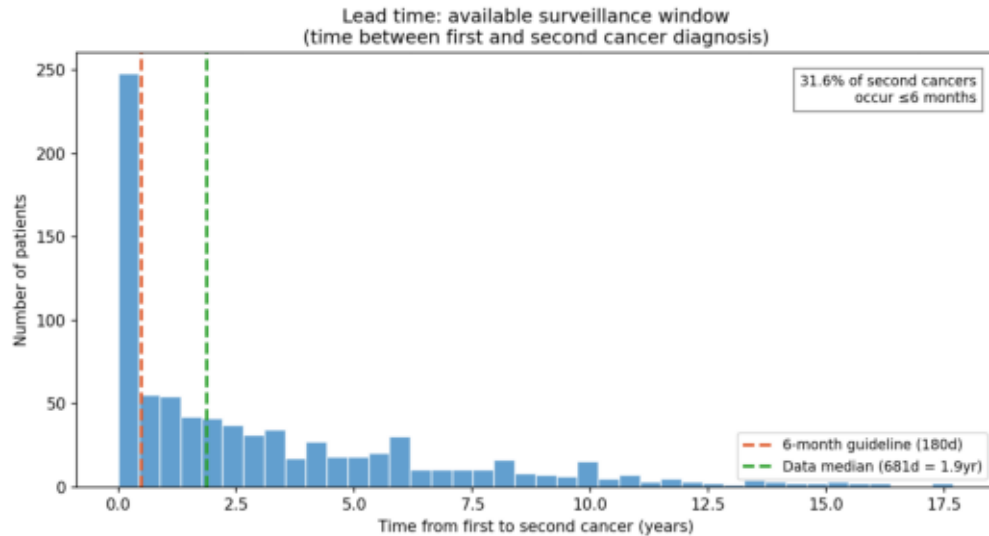
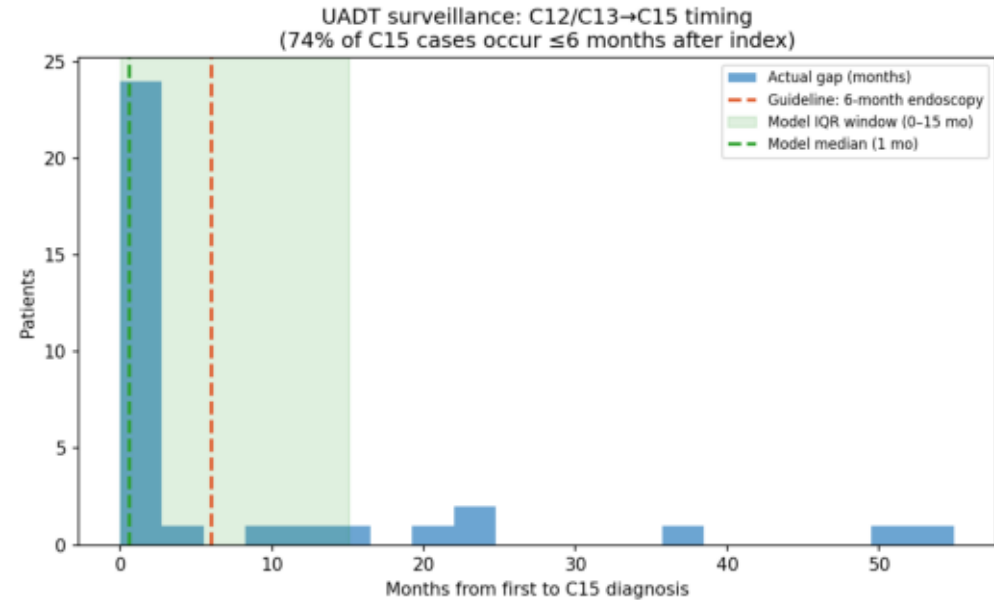
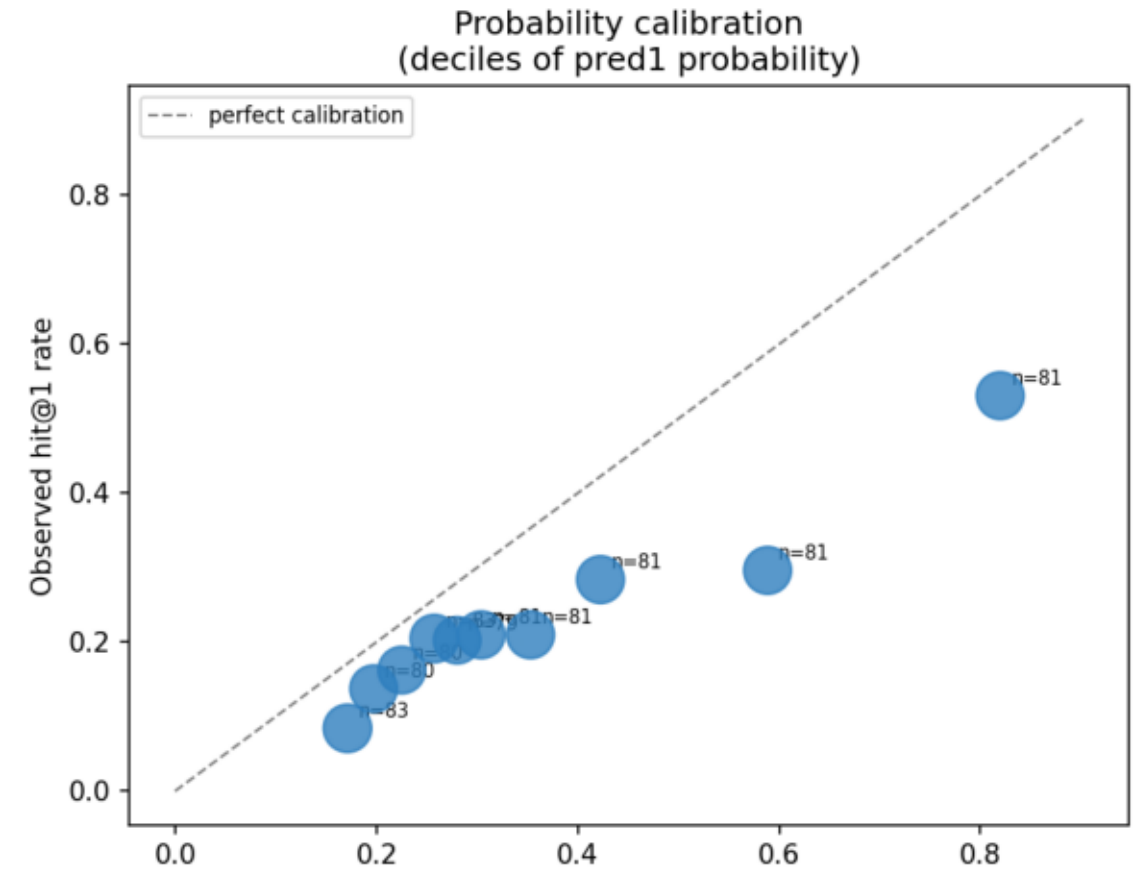


Fig 2b: UADT timing — C12/C13→C15
vs 6-month guideline (red dashed)



Probability Calibration

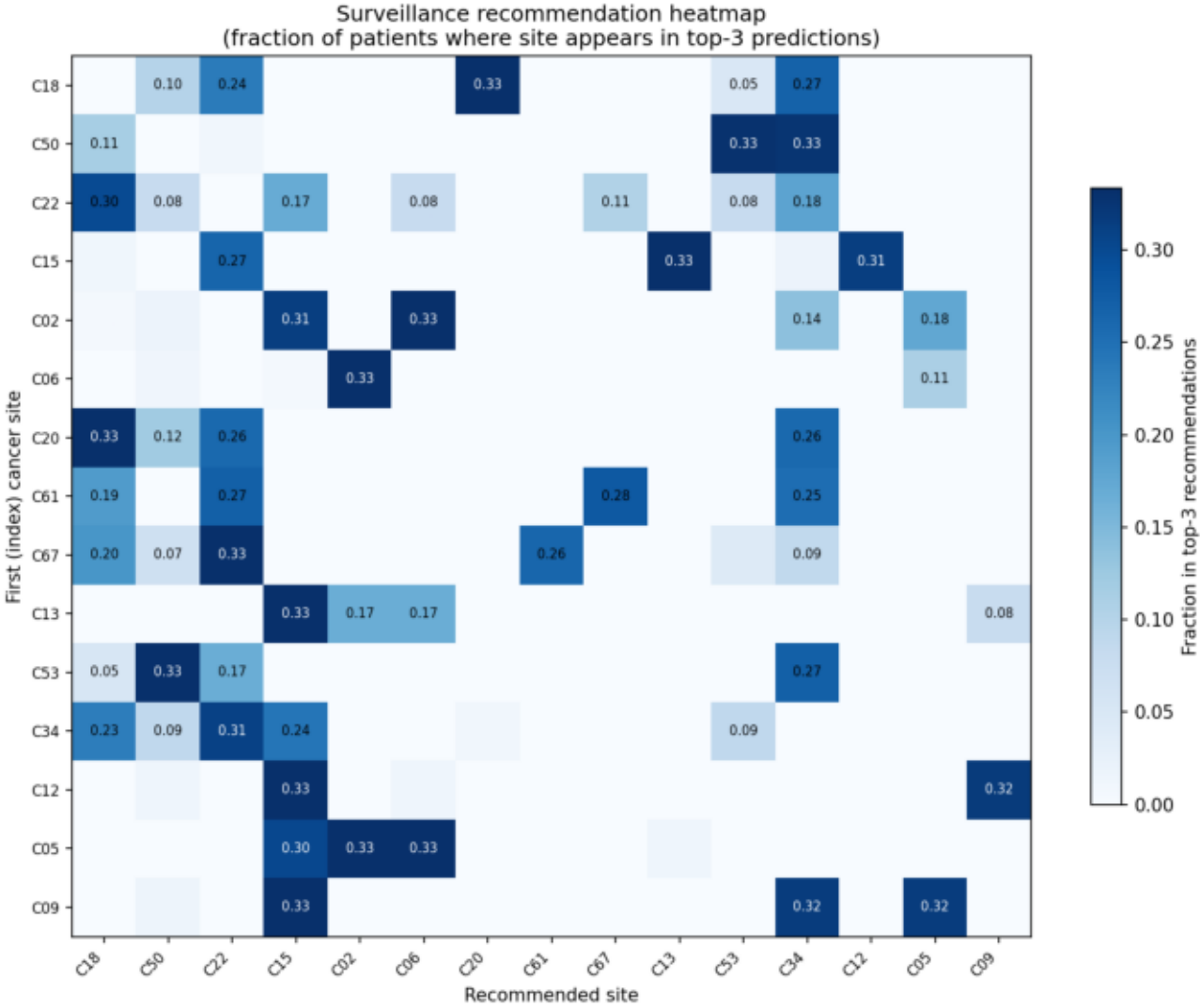
Fig 3: Probability calibration — model prob1 vs observed R@1
(bubble size = n patients in decile)



Note: all 810 validation patients received prob1 > 0.15 (model is uniformly high-confidence). This reflects the small vocabulary (~37 sites): after zeroing special tokens and the primary site, probability concentrates on ~35 targets. Risk stratification should use relative thresholds (e.g., tertiles of prob1) rather than absolute cutoffs. The calibration plot shows whether the model's relative confidence tracks observed accuracy.

Site-to-Site Recommendation Map

Fig 4: Surveillance recommendation heatmap
(for each first-cancer site, fraction of patients where target appears in top-3)



Surveillance Timing Windows

Table 1. Empirical surveillance timing windows
(top 20 pairs by N, ≥10 observed transitions)

Index site	Target site	N pairs	Q1 (months)	Median (months)	Q3 (months)
C50	C34	81	18.0	49.3	114.7
C18	C20	69	0.0	0.0	3.5
C13	C15	62	0.1	0.9	24.9
C18	C22	59	4.7	35.0	73.4
C02	C06	58	1.4	33.2	64.5
C12	C15	57	0.1	0.4	5.3
C53	C50	57	23.4	66.2	129.2
C22	C34	53	7.1	24.0	62.7
C18	C61	52	14.8	47.1	73.4
C18	C34	51	7.7	28.7	79.5
C15	C13	50	0.5	1.9	19.2
C67	C61	50	2.0	5.1	19.6
C50	C18	48	21.6	53.6	86.1
C06	C02	44	24.0	44.1	66.6
C50	C54	44	39.3	67.6	108.3
C03	C06	43	0.6	23.9	48.1
C50	C22	42	18.5	47.5	89.0
C06	C03	42	31.0	46.4	70.7
C02	C03	40	11.7	39.0	68.7
C61	C22	35	2.3	18.3	53.4

Timing window interpretation: 'Recommend screening for [Target] starting at [Q1] months and continuing through [Q3] months after [Index] diagnosis.' Pairs with Q1≈0 include synchronous presentations (≤6 months) — these should trigger immediate workup at the time of index diagnosis.

Clinical Interpretation + Limitations + Next Steps

Key findings:

1. First-cancer-only $R@1=0.232$ ($8.6\times$ random chance, $1/37=0.027$). The Transformer has learned site-level co-occurrence priors that generalise to unseen patients.
2. Median inter-cancer gap = 681d (1.9yr), IQR 48–1813d.
This is the window in which surveillance has an opportunity to act. Most second cancers occur beyond the 6-month guideline window — supporting extended protocols.
3. UADT validation: 32% of C12/C13→C15 transitions occur ≤ 6 months.
The model's 0–15 month timing window encompasses this high-density early window and extends it, capturing the 26% of late-presenting cases.
4. The recommendation heatmap reveals non-UADT pairs of clinical interest: liver (C22) appears in top-3 for multiple index sites — consistent with the HBV pan-carcinogen signal identified in the masked predictor (Script 02b).

Limitations:

- ① First-cancer-only context does not use temporal history of ≥ 2 prior cancers — $R@1$ would increase to 0.312 with full history (Script 07).
- ② Timing windows derived from observed transitions, not from the model — the Transformer does not yet directly predict time-to-event.
- ③ Validation is retrospective; prospective RCT or quasi-experimental design with EHR integration required to demonstrate clinical utility.
- ④ Risk calibration: all patients receive high-confidence predictions ($\text{prob1}>0.15$) due to small vocab; relative tertile thresholds recommended.

Next steps:

- Extend Transformer to jointly predict site AND time-to-event (add regression head).
- Prospective validation: embed calendar into EHR; compare endoscopy yield vs standard-of-care over 2 years.
- Combine cluster membership (Script 10) as a conditioning token to personalise recommendations by carcinogenic axis.